

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
 Manufacturer: Barco
 Firmware Version: N/A
 Model(s): F90-4K13, F90-W13, F80-Q9

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.1

Version History

Module Version	Date	Notes
1_0_3_0	12/20/2017	Added model F80-Q9
1_0_2_0	2/13/2017	Fixed all commands
1_0_0_0	12/27/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='F90-4K13')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='F90-4K13')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9090, Model='F90-4K13')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command Focus	Value 'In'	Value 'Out'	
Qualifier Key 'Steps'	Qualifier Value '1' – '16'		
# Focus example InterfaceName.Set('Focus', 'In', {'Steps': '1'})			
Command Illumination	Value 0 to 100 in steps of 1		
# Illumination example InterfaceName.Set('Illumination', 100)			
Command Input	Value 'DVI 1' 'DisplayPort 2' 'SDI'	Value 'DVI 2' 'HDMI'	Value 'DisplayPort 1' 'HDBaseT'
# Input example InterfaceName.Set('Input', 'DVI 1')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Shutter	Value 'Open'	Value 'Closed'	
# Shutter example InterfaceName.Set('Shutter', 'Open')			
Command TestPattern	Value 'On'	Value 'Off'	
# TestPattern example InterfaceName.Set('TestPattern', 'On')			
Command TestPatternSelect	Value 'Focus Black' 'Checker Board' 'Monoscope'	Value 'Ansi Lumen' 'Focus' 'Outline'	Value 'Scenergix' 'Cross Hatch' 'Color Bars'
# TestPatternSelect example InterfaceName.Set('TestPatternSelect', 'Focus Black')			

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Command Zoom	Value 'In'	Value 'Out'
Qualifier Key 'Steps'	Qualifier Value '1' – '16'	
# Zoom example InterfaceName.Set('Zoom', 'In', {'Steps': '1'})		

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value		
Illumination	0 to 100 in steps of 1		
# Illumination examples InterfaceName.Update('Illumination') Value = InterfaceName.ReadStatus('Illumination') InterfaceName.SubscribeStatus('Illumination', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'DVI 1'	'DVI 2'	'DisplayPort 1'
	'DisplayPort 2'	'HDMI'	'HDBaseT'
	'SDI'		
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	
Power	'On'	'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command	Value	Value	
Shutter	'Open'	'Closed'	
# Shutter examples InterfaceName.Update('Shutter')			

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<pre>Value = InterfaceName.ReadStatus('Shutter') InterfaceName.SubscribeStatus('Shutter', None, FeedbackHandler)</pre>			
Command TestPattern	Value 'On'	Value 'Off'	
<pre># TestPattern examples InterfaceName.Update('TestPattern') Value = InterfaceName.ReadStatus('TestPattern') InterfaceName.SubscribeStatus('TestPattern', None, FeedbackHandler)</pre>			
Command TestPatternSelect	Value 'Focus Black' 'Checker Board' 'Monoscope'	Value 'Ansi Lumen' 'Focus' 'Outline'	Value 'Scenergix' 'Cross Hatch' 'Color Bars'
<pre># TestPatternSelect examples InterfaceName.Update('TestPatternSelect') Value = InterfaceName.ReadStatus('TestPatternSelect') InterfaceName.SubscribeStatus('TestPatternSelect', None, FeedbackHandler)</pre>			

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

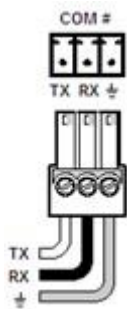
Data Bits: 8

Parity: None

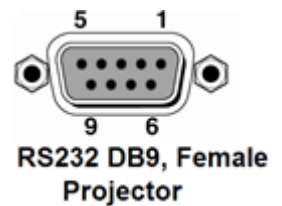
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
RxD	←	2	TxD
TxD	→	3	RxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	9090
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Focus In Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepforward", "steps": 1}, "id": 1}\x0D\x0A
Focus Out Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepreverse", "steps": 1}, "id": 1}\x0D\x0A
Focus In Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepforward", "steps": 16}, "id": 1}\x0D\x0A
Focus Out Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepreverse", "steps": 16}, "id": 1}\x0D\x0A
Illumination 0	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 0.0}, "id": 2}\x0D\x0A
Illumination 100	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 100.0}, "id": 2}\x0D\x0A
Input DVI 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DVI 1"}, "id": 5}\x0D\x0A
Input DVI 2	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DVI 2"}, "id": 5}\x0D\x0A
Input DisplayPort 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DisplayPort 1"}, "id": 5}\x0D\x0A
Input DisplayPort 2	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DisplayPort 2"}, "id": 5}\x0D\x0A
Input HDBaseT	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "HDBaseT"}, "id": 5}\x0D\x0A
Input HDMI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "HDMI"}, "id": 5}\x0D\x0A
Input SDI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "SDI"}, "id": 5}\x0D\x0A

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	"image.window.main.source", "value": "SDI"},"id": 5}\x0D\x0A
Power Off	{"jsonrpc": "2.0", "method": "system.poweroff", "params": {}, "id": 3}\x0D\x0A
Power On	{"jsonrpc": "2.0", "method": "system.poweron", "params": {}, "id": 3}\x0D\x0A
Shutter Closed	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.shutter.target", "value": "Closed"}, "id": 4}\x0D\x0A
Shutter Open	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.shutter.target", "value": "Open"}, "id": 4}\x0D\x0A
Test Pattern Off	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.show", "value": false}, "id": 6}\x0D\x0A
Test Pattern On	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.show", "value": true}, "id": 6}\x0D\x0A
Test Pattern Select Ansi Lumen	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Ansi Lumen"}, "id": 7}\x0D\x0A
Test Pattern Select Checker Board	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Checker Board"}, "id": 7}\x0D\x0A
Test Pattern Select Color Bars	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Color Bars"}, "id": 7}\x0D\x0A
Test Pattern Select Cross Hatch	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Cross Hatch"}, "id": 7}\x0D\x0A
Test Pattern Select Focus	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Focus"}, "id": 7}\x0D\x0A
Test Pattern Select Focus Black	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Focus Black"}, "id": 7}\x0D\x0A
Test Pattern Select Monoscope	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Monoscope"}, "id": 7}\x0D\x0A
Test Pattern Select Outline	{"jsonrpc": "2.0", "method": "property.set", "params": {"property":

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	"image.testpattern.selected", "value": "internal:Outline"},"id": 7}\x0D\x0A
Test Pattern Select Scenergix	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Scenergix"},"id": 7}\x0D\x0A
Zoom In Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepforward", "steps": 1}, "id": 8}\x0D\x0A
Zoom Out Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepreverse", "steps": 1}, "id": 8}\x0D\x0A
Zoom In Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepforward", "steps": 16}, "id": 8}\x0D\x0A
Zoom Out Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepreverse", "steps": 16}, "id": 8}\x0D\x0A

Appendix B. Update Commands

Illumination	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "illumination.sources.laser.power"}, "id": 2}\x0D\x0A</code>
Input	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.window.main.source"}, "id": 5}\x0D\x0A</code>
Power	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "system.state"}, "id": 3}\x0D\x0A</code>
Shutter	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "optics.shutter.target"}, "id": 4}\x0D\x0A</code>
Test Pattern Select	<code>{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.testpattern.selected"}, "id": 7}\x0D\x0A</code>